Garbage-Guard Field Kit · Before and after each field day, Sheet 1: Uptime Log (Table 16)

Print, bring to the creek and fill in by hand. Everything here feeds Chapter IV. Take a phone photo of each filled page at the end of the day.

Before each field day

☐ Charger, extension lead and USB Wi-Fi adapter packed
☐ Camera clock set (camera settings, System, Time Settings, Sync with computer time)
☐ Settings checked and written down: model _____ detection size _____ stream _____ threshold _____
☐ Monitored area drawn tightly around the water, with the floating barrier and the banks outside it
☐ Remote screen sharing turned off (it slows the Pi down and spoils the latency numbers)

At the end of each field day

Run these on the Pi, then copy both files to a PC. Do not use Delete in the History tab during the field week.

```
cp ~/gg/garbageguard.db ~/gg_backup_$(date +%F).db
tar czf ~/snapshots_$(date +%F).tgz -C ~/gg snapshots
```

Sheet 1: Uptime Log (Table 16)

Date	Start time	Stop time	Hours	Model / size	Threshold	Weather and light	Crash, freeze or overheating? What happened?

Garbage-Guard Field Kit · Sheet 2: Threshold Calibration Record (Table 17)

Count the items inside the monitored area by eye, then read the count on the dashboard at the same moment. Write the threshold before and after any change.

Date and time	Hand count in the monitored area	Dashboard count	Difference	Threshold before	Threshold after	Notes

Garbage-Guard Field Kit · Sheet 3: Test Case Trials (Tables 14, 15 and 19)

Do each test 10 times. Tick a box for a pass, cross it for a fail, and write what happened on a fail in Notes.

Test	What to do	What counts as a pass	Trials 1 to 10	Passed / 10	Notes
TC_UNIT_02	On the Pi: `python3 test_counter.py`, plus one simulated count up to the threshold	Alert once at the threshold, no second alert until the count falls to the re-arm level	☐☐☐☐☐☐☐☐☐☐		
TC_UNIT_03	Play a recorded video through the Pi	Latency is logged and nothing crashes	☐☐☐☐☐☐☐☐☐☐		
TC_SYS_01	Watch the live feed with real floating waste	Boxes on visible bottles, bags and containers; nothing on sunken items or leaves	☐☐☐☐☐☐☐☐☐☐		
TC_SYS_02	Let items build up past the threshold	Accumulation Alert is raised	☐☐☐☐☐☐☐☐☐☐		
TC_SYS_03	Run for a set time (write it down)	Latency stays steady, no crash or overheating	☐☐☐☐☐☐☐☐☐☐		
TC_SYS_04	Full run: switch on, feed appears, add items, count rises, alert, record saved, app notified	Every step happens in order (take a photo of each step once)	☐☐☐☐☐☐☐☐☐☐		
TC_SYS_05	Open the Monitor tab while detecting	Live boxes, labels, count and frame rate keep updating	☐☐☐☐☐☐☐☐☐☐		
TC_SYS_06	Put the same item inside, then outside, the area	Counted inside, ignored outside	☐☐☐☐☐☐☐☐☐☐		
TC_SYS_07	Let one item drift in and out of the area	Same ID number the whole time; count goes up one, then back down	☐☐☐☐☐☐☐☐☐☐		
TC_SYS_08	Raise an alert, take a Capture, open History	Both records there with time, counts and snapshot	☐☐☐☐☐☐☐☐☐☐		
TC_SYS_09	Raise an alert with the phone app connected	Dashboard shows it, buzzer sounds, phone gets a notification	☐☐☐☐☐☐☐☐☐☐		
TC_SYS_10	Change the threshold in Settings, repeat the count	Alert comes at the new value	☐☐☐☐☐☐☐☐☐☐		
TC_ACC_01	Count correct labels against all items seen, on several snapshots	Write correct / total; the target is 80%	Correct ____ / Total ____	____%	
TC_ACC_02	Count alerts raised against threshold crossings you saw	The target is 80%	Raised ____ / Crossings ____	____%	
TC_ACC_03	Read the latency records afterwards (done on the PC)	Average under 1.5 s, no crash	–	–	

Garbage-Guard Field Kit · Sheet 4: Usability Survey (Table 20)

For barangay personnel, after they have used the touchscreen and the phone app. One sheet per person.

Name (optional): _____ Position: _____ Date: _____

Encircle one number for each statement: 5 Strongly Agree 4 Agree 3 Neutral 2 Disagree 1 Strongly Disagree

Code	Statement	Rating
USR-U1	Information displayed on the monitor is easy to read and understand.	5 4 3 2 1
USR-U2	Visual bounding boxes on the camera feed clearly showed where the garbage is located on the creek.	5 4 3 2 1
USR-U3	The Accumulation Alert is clear, noticeable and effectively informs when the creek needs cleaning.	5 4 3 2 1
USR-U4	The system provides helpful insights that would assist local authorities in monitoring creek pollution.	5 4 3 2 1
USR-U5	The Android application shows the alerts and the history clearly, and what it shows matches the touchscreen.	5 4 3 2 1

Comments or suggestions:

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